

Geometric Structure in Neural Population Codes

A tool-comparison framework for causal neural geometry

Exploratory results on Steinmetz 2019

Elliott Tower

2026

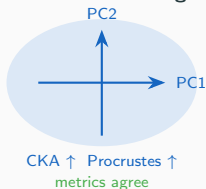
Two tools, opposite answers

Two standard methods for measuring neural similarity—CKA (linear kernel alignment) and Procrustes (manifold shape distance)—are routinely used interchangeably. We show they systematically disagree, and that effective dimensionality explains why.

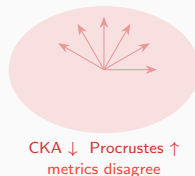
This matters: which metric you use determines what you conclude about which brain regions “match.”

We then test whether the geometry is causally meaningful using interchange interventions, validated against criteria from philosophy of science.

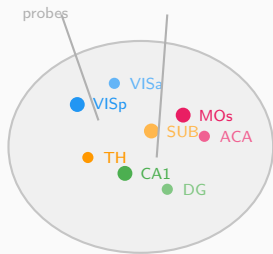
Low-dimensional region



High-dimensional region



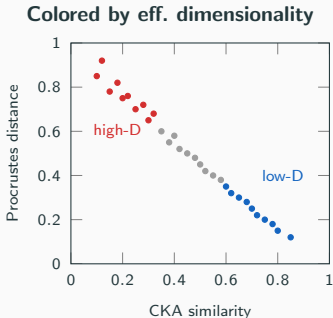
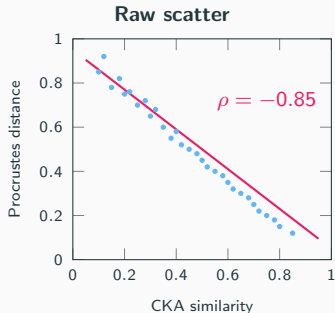
The data: Steinmetz et al. 2019



- 39 sessions, 10 mice
- **73 brain regions** recorded simultaneously
- Neuropixels probes (high density)
- Visual discrimination task (which side has higher contrast?)
- 39 sessions $\times$ $\sim$ 73 regions = **1,316 pairwise comparisons** for metric analysis

Steinmetz et al., Nature 2019

CKA and Procrustes systematically disagree



The anti-correlation ($\rho = -0.85$) is mediated by effective dimensionality.

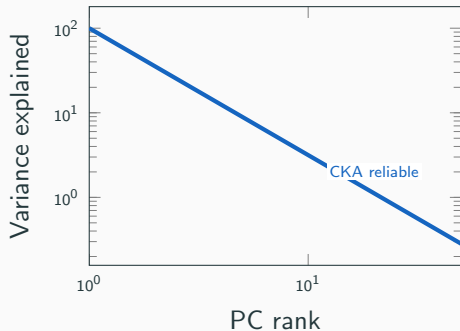
Partial ρ controlling for dimensionality: -0.31 .

High-D regions look “different” to CKA and “similar” to Procrustes; low-D regions flip.

Replication: held from 5 $\rightarrow$ 39 sessions

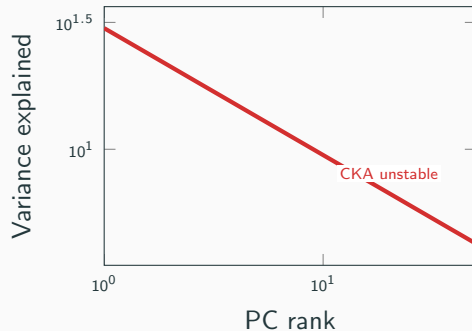
Why CKA and Procrustes disagree

Steep spectrum (VISp-type)



Low-D: dominant eigenvectors stable across animals

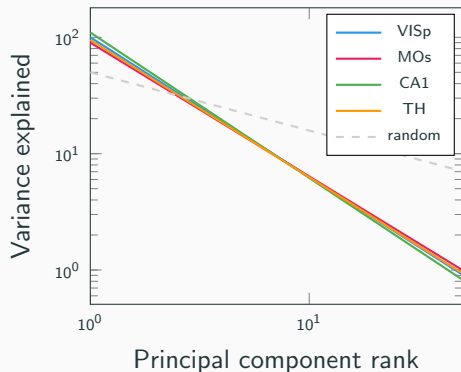
Flat spectrum (MOs-type)



High-D: no dominant direction $\rightarrow$ CKA captures noise

CKA is sensitive to the dominant eigenvectors. In high-D regions, no direction dominates, so CKA is unstable across animals. Procrustes compares the full k -dimensional subspace and is stable even when individual directions differ.

Eigenvalue shapes are universal; trial content is not



Eigenvalue shape: $r = 0.936$ across regions

Power-law decay is near-identical everywhere

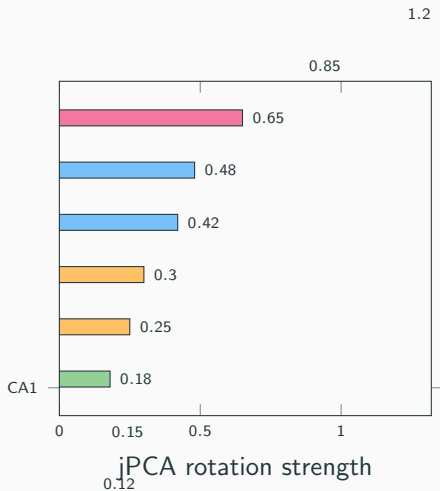
Trial-specific content: $r = 0.089$

Cross-animal similarity of PC projections is low (comparable to permuted-trial null)

The scaffolding is shared; what gets built on it is region-specific.

Subheading: eigenvalue *shape* is universal, but the *modes* are idiosyncratic.

Every region rotates (but at different speeds)



72/73 regions above shuffle null

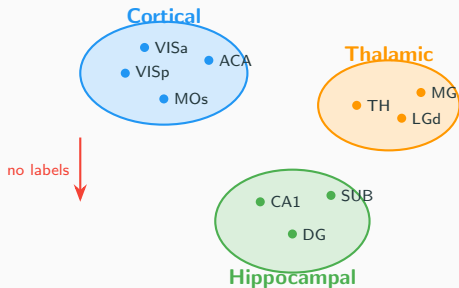
10× variation in rotation strength

Frontal/motor	strongest
Sensory	moderate
Thalamic	moderate
Hippocampal	weakest

Rotational dynamics are ubiquitous but not uniform. The hierarchy is interpretable: motor planning regions rotate fast; memory regions rotate slowly.

jPCA, Churchland et al. 2012

Representational similarity matches macro-anatomy



Input: CKA similarity matrix (no anatomy)

Output: $k = 3$ clusters

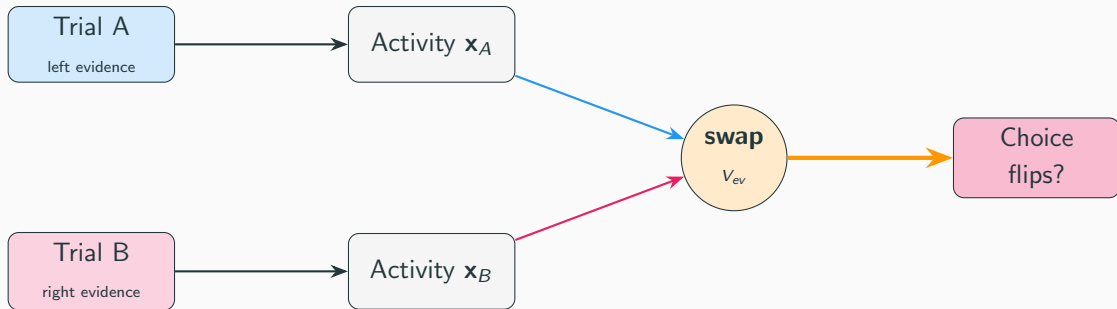
Result: matches cortical / thalamic / hippocampal

Regions with distinct firing statistics and temporal dynamics cluster appropriately. This is **expected**—but useful as a sanity check that our metric captures biological structure, not noise.

Replicates at session level and cross-animal.

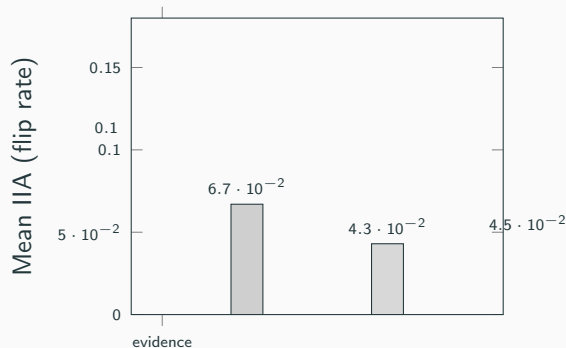
Causal interventions

Interchange Intervention Analysis (IIA)



Swap the “choice-relevant dimensions” of neural activity between two trials $\rightarrow$ if the brain’s prediction flips, that subspace causally controls choice.
(Adapted from DAS, Geiger et al. 2024, applied to biological recordings.)

IIA: swapping evidence flips choices



Absolute effect: $\sim 10\%$ flip rate

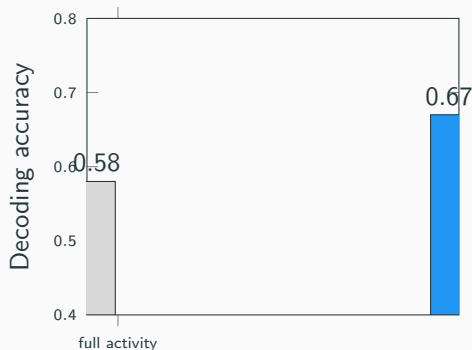
Random baseline: $\sim 4.5\%$ flip rate

Specificity ratio: $2.3\times$

The effect is modest in absolute terms—but the evidence axis produces more than double the flip rate of random or feedback axes, with $p < 0.001$ (Mann-Whitney).

Swapping other axes (RT, feedback, random) produces flip rates near the random baseline. The evidence subspace has a specific causal role.

The evidence subspace is sufficient (and denoises)



Recovery fraction = 1.16

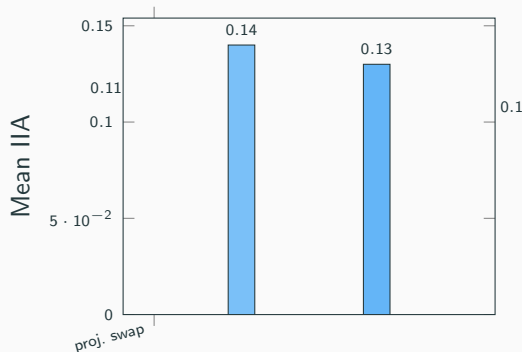
The evidence subspace alone decodes choice *better* than full activity—it removes noise dimensions that hurt the classifier.

20/24 regions pass (≥ 0.70 recovery)

4 failures are all high-dimensional (flat-spectrum) regions where subspace estimation is harder.

Sufficiency + necessity (IIA) together establish the evidence subspace as a causal mediator.

Four intervention methods agree



Methods:

1. **Projection swap**: replace V -projected component
2. **Gaussian noise**: add noise along V
3. **Mean-shift**: shift V -projected mean to opposite class
4. **Zeroing**: zero out V -projected component

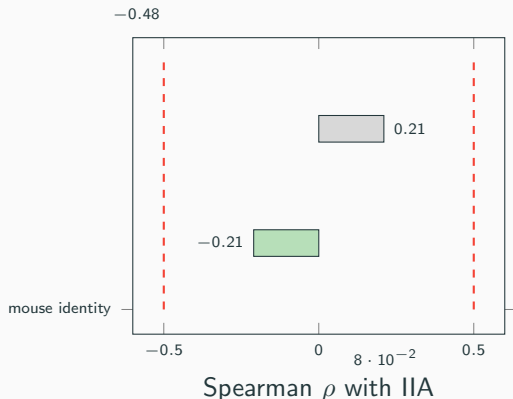
Cross-method rank correlation:

$$\bar{\rho} = 0.72$$

Regions ranking is preserved across methods.

The effect is not an artifact of any one intervention type.

Confound control



Dashed lines = $|\rho| > 0.5$ threshold

No confound exceeds $|\rho| = 0.5$

- Firing rate: $\rho = 0.21$ (not significant)
- Trial count: $\rho = -0.21$ (not significant)
- Mouse identity: $\rho = 0.08$ (effect holds within-mouse)

Caveat: neuron count shows $\rho = -0.48$ —below threshold but a near-miss. Addressed in limitations.

exp51

Novel predictions and honest nulls

Geometric type predicts decoder choice

Regions where CKA is high (linear-type) favor LDA; regions where Procrustes is high (manifold-type) favor kNN. An out-of-sample prediction: geometric type determines which decoder works, without training on decoding data.

$$\rho = -0.509, p = 0.00016 \ (n = 50 \text{ regions})$$

Causal abstraction geometry

Power-law exponent α correlates with evidence—choice subspace routing.

$$\rho = 0.288, p = 0.014 \ (\text{moderate})$$

Allen Atlas: honest null

Structural connectivity from the Allen Mouse Brain Atlas does not predict functional IIA.

$$\rho = 0.031, p = 0.65$$

Anatomical wiring does not predict which regions causally route evidence. The functional geometry we found is not recapitulating known anatomy—it is capturing something different.

IBL cross-dataset: replication in progress (exp46). Results pending.

Criteria adapted from Craver (2007), Shallice (1988), Mayo (2018). We report failures alongside successes.

What this means

Summary

What we found:

- CKA and Procrustes systematically disagree ($\rho = -0.85$); effective dimensionality explains why
- Rotational dynamics are widespread (72/73 regions) but not uniform ($10\times$ variation)
- The evidence subspace causally mediates choice (IIA, 4 methods, sufficiency)
- Representational geometry tracks anatomy at the macroscale

What this means:

Neural geometry is **not metric-neutral**. The tool you choose determines what you find—and dimensionality tells you which tool to use. The causal structure we identify is modest in effect size ($\sim 10\%$ flip rate) but robust across methods and specific to the evidence axis.

Limitations and next steps

Honest limitations:

- One dataset (Steinmetz 2019)
- Binary task only—unclear if this generalizes
- IIA effect sizes are modest ($\sim 10\%$ flip rate, $\sim 4.5\%$ baseline)
- Double dissociation is partial (18% rate)
- Neuron count $\rho = -0.48$ is a near-miss confound
- Exploratory analysis; hypotheses for IBL replication now specified

Next:

- IBL cross-dataset replication (in progress)
- Zatz-Haas optogenetic validation (silencing data downloaded)
- Multi-task generalization
- Dimensionality correction for neuron count confound

Intellectual grounding

This work draws on ideas from three research traditions:

- **Causal representation learning** (Siddharth Mishra-Sharma) — identifiability conditions for when geometric structure in latent spaces reflects true causal variables
- **Relational causal models** (Jensen & Neville) — extending causal inference beyond i.i.d. data to relational and multi-level systems, which is what brain region interactions are
- **Dynamical systems in neuroscience** (Hava Siegelmann) — neural computation as geometry of dynamical trajectories; connections between topology and computational capacity

Validity framework grounded in philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003).

Code & results: [ZENODO DOI / GITHUB URL]

Neural geometry is not metric-neutral.

**The tool determines the finding—dimensionality tells you
which tool to use.**